

Tugas 4 – Perbaikan Citra

Diketahui bitmap 24 bit dalam nilai integer sbb:

\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4
\$001267F4	\$001267F4	\$004F20FF	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$00FF005A	\$001267F4
\$001267F4	\$001267F4	\$004F20FF	\$004F20FF	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$00FF005A	\$00FF005A	\$001267F4
\$001267F4	\$001267F4	\$004F20FF	\$004F20FF	\$004F20FF	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$00FF005A	\$00FF005A	\$00FF005A	\$001267F4
\$001267F4	\$001267F4	\$004F20FF	\$004F20FF	\$004F20FF	\$004F20FF	\$001267F4	\$001267F4	\$00FF005A	\$00FF005A	\$00FF005A	\$00FF005A	\$001267F4
\$001267F4	\$001267F4	\$004F20FF	\$004F20FF	\$004F20FF	\$004F20FF	\$004F20FF	\$004F20FF	\$00FF005A	\$00FF005A	\$00FF005A	\$00FF005A	\$001267F4
\$001267F4	\$001267F4	\$004F20FF	\$004F20FF	\$004F20FF	\$004F20FF	\$001267F4	\$001267F4	\$00FF005A	\$00FF005A	\$00FF005A	\$00FF005A	\$001267F4
\$001267F4	\$001267F4	\$004F20FF	\$004F20FF	\$004F20FF	\$001267F4	\$001267F4	\$001267F4	\$00FF005A	\$00FF005A	\$00FF005A	\$00FF005A	\$001267F4
\$001267F4	\$001267F4	\$004F20FF	\$004F20FF	\$004F20FF	\$001267F4	\$001267F4	\$001267F4	\$00FF005A	\$00FF005A	\$00FF005A	\$00FF005A	\$001267F4
\$001267F4	\$001267F4	\$004F20FF	\$004F20FF	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$00FF005A	\$00FF005A	\$00FF005A	\$00FF005A	\$001267F4
\$001267F4	\$001267F4	\$004F20FF	\$004F20FF	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$00FF005A	\$00FF005A	\$00FF005A	\$00FF005A	\$001267F4
\$001267F4	\$001267F4	\$004F20FF	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$00FF005A	\$00FF005A	\$00FF005A	\$00FF005A	\$001267F4
\$001267F4	\$001267F4	\$004F20FF	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$00FF005A	\$00FF005A	\$00FF005A	\$00FF005A	\$001267F4
\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4	\$001267F4

1. Lakukan proses Brightness terhadap masing-masing channel RGB nya, tampilkan hasilnya dalam bentuk matriks bitmap pada masing-masing channel

Jawaban:

- Matriks Channel R (Original):
Kebanyakan bernilai 18, blok tengah kiri 79, blok kanan bawah 255.
- Matriks Channel G (Original):
Kebanyakan bernilai 103, blok tengah kiri 32, blok kanan bawah 0.
- Matriks Channel B (Original):
Kebanyakan bernilai 244, blok tengah kiri 255, blok kanan bawah 90.

2. Hitung inversnya pada masing-masing channel RGB nya dan pada citra grayscalenya

- \$001267F4:
 $R = 255 - 18 = 237$
 $G = 255 - 103 = 152$
 $B = 255 - 244 = 11$
- \$004F20FF:
 $R = 255 - 79 = 176$
 $G = 255 - 32 = 223$
 $B = 255 - 255 = 0$
- \$00FF005A:
 $R = 255 - 255 = 0$
 $G = 255 - 0 = 255$
 $B = 255 - 90 = 165$

RGB berdasarkan channel:

- \$001267F4:
 $(0.299 * 18) + (0.587 * 103) + (0.114 * 244) = 93.65$
 $255 - 93.65 = 161.35$

- \$004F20FF:
 $(0.299 * 79) + (0.587 * 32) + (0.114 * 255) = 71.47$
 $255 - 71.47 = 183.53$
- \$00FF005A:
 $(0.299 * 255) + (0.587 * 0) + (0.114 * 90) = 86.51$
 $255 - 86.51 = 168.49$

3. Hitung peregangan contrast pada citra grayscalenya dengan pusat kontras P= 100 dan penguatan kontras G = 3
4. Buatlah program untuk proses brightness dengan input nilai brightness k

```
import numpy as np

def proses_brightness(matriks, k):
    hasil = np.clip(matriks.astype(np.int16) + k, 0, 255)
    return hasil.astype(np.uint8)

matriks_soal = np.array([
    [[18, 103, 244], [18, 103, 244]],
    [[79, 32, 255], [79, 32, 255]],
    [[255, 0, 90], [255, 0, 90]]
], dtype=np.uint8)

k = 50
hasil = proses_brightness(matriks_soal, k)

print(f"--- HASIL BRIGHTNESS (k={k}) ---")
print(hasil)
```

5. Buatlah program untuk proses invers

```
import numpy as np

def proses_invers(matriks):
    return (255 - matriks).astype(np.uint8)

matriks_soal = np.array([
    [[18, 103, 244],
     [79, 32, 255],
     [255, 0, 90]]
], dtype=np.uint8)

hasil = proses_invers(matriks_soal)

print("--- HASIL INVERS ---")
```

```
print(hasil)
```

6. Buatlah program untuk proses peregangkan kontras dengan input Pusat Kontras P dan Penguatan kontras G

```
import numpy as np

def proses_kontras(matriks_gray, P, G):
    img_float = matriks_gray.astype(np.float32)
    hasil = P + G * (img_float - P)
    return np.clip(hasil, 0, 255).astype(np.uint8)

rgb_data = np.array([[18, 103, 244], [79, 32, 255], [255, 0, 90]],
dtype=np.uint8)
gray_soal = (0.299*rgb_data[:,0] + 0.587*rgb_data[:,1] +
0.114*rgb_data[:,2]).astype(np.uint8)

P_input = 100
G_input = 3
hasil = proses_kontras(gray_soal, P_input, G_input)

print(f"--- HASIL CONTRAST STRETCHING (P={P_input}, G={G_input}) ---")
print("Nilai Gray Asli:", gray_soal)
print("Nilai Gray Hasil Kontras:", hasil)
```